

Tilting étale cohomology of perfectoid towers

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§ 1. Introduction

$$\begin{array}{ccc} A = \varprojlim \mathbb{Z}_p[p^{\frac{1}{p^n}}] & \xrightarrow{\text{tilting}} & \varprojlim \mathbb{F}_p[[T]][T^{\frac{1}{p^n}}] = A^b \\ \cup & & \cup \\ \vdots & & \vdots \\ \cup & & \cup \\ \mathbb{Z}_p[p^{\frac{1}{p^2}}] & & \mathbb{F}_p[[T]][T^{\frac{1}{p^2}}] \\ \cup & & \cup \\ \mathbb{Z}_p[p^{\frac{1}{p}}] & & \mathbb{F}_p[[T]][T^{\frac{1}{p}}] \\ \cup & & \cup \\ R = \mathbb{Z}_p & \xleftrightarrow{\text{analogy}} & \mathbb{F}_p[[T]] = R^{s.b} \\ \cup & & \cup \\ P & \xleftrightarrow{\quad} & T = p^b = p^{s.b} \end{array}$$

Thm 1.1 (Česnavičius-Scholze '24)

A : a perfectoid ring

$$\begin{array}{ccc}
 \text{Spec}(A) & & \text{Spec}(A^b) \\
 \cup & & \cup \\
 \overline{U} & \longleftrightarrow & \overline{U}^b \\
 \cup & & \cup \\
 \text{Spec}(A) \setminus V(p) & & \text{Spec}(A^b) \setminus V(p^b) \\
 & \xrightarrow{\cong} &
 \end{array}$$

$$\Rightarrow \mathbb{R}\Gamma_{\text{ét}}(U, G) \cong \mathbb{R}\Gamma_{\text{ét}}(U^b, G)$$

for $\forall G$: torsion abel. gp.

Thm 1.2 (H.)

$R = R_0 \rightarrow R_1 \rightarrow R_2 \rightarrow \dots$: a perfectoid tower

s.t. $R_i \rightarrow R_{i+1}$ are integral inj.

$$\begin{array}{ccc}
 \text{Spec}(R) & & \text{Spec}(R^{s.b}) \\
 \cup & & \cup \\
 \overline{U} & \longleftrightarrow & \overline{U}^{s.b} \\
 \cup & & \cup \\
 \text{Spec}(R) \setminus V(p) & & \text{Spec}(R^{s.b}) \setminus V(p^{s.b})
 \end{array}$$

$$\Rightarrow \mathbb{R}\Gamma_{\text{ét}}(U, G) \cong \mathbb{R}\Gamma_{\text{ét}}(U^b, G)$$

for $\forall G$: torsion abel. gp.

Rem 1.3 Ishino-Nakazato-Shimomoto already proved

$$\# H_{\text{ét}}^{\mathbb{Q}}(U, G) \leq \# H_{\text{ét}}^{\mathbb{Q}}(U^{\text{s.b.}}, G)$$

under suitable normality assumptions.

§2. Perfectoid towers

- ring = commutative
- Fix a prime number p .

Def 2.1 A ring A is **perfectoid**

$(\Leftrightarrow)_{\text{def}} \exists \varpi \in A$ with $p \in \varpi^p A$ s.t.

① A is ϖ -adically complete.

$$\textcircled{2} \begin{array}{ccc} A/\varpi^p & \xrightarrow{\varphi} & A/\varpi^p \\ \downarrow & \curvearrowright & \vdots \\ A/\varpi & \xrightarrow{\varphi} & A/\varpi \end{array}$$

$$\textcircled{3} \begin{array}{ccc} A[\varpi^\infty] & \xrightarrow{\varphi} & A[\varpi^\infty] \\ \downarrow & & \downarrow \\ X & \xrightarrow{\varphi} & X^p \end{array}$$

- $A^b \stackrel{\text{def}}{=} \varprojlim (A/p \xleftarrow{\varphi} A/p \xleftarrow{\varphi} A/p \xleftarrow{\varphi} \dots)$: **tilt** perfect(oid)

$\rightarrow \text{pr}_0 : A^b \rightarrow A/p$ induces $A^b_{p^b} \xrightarrow{\varphi} A/p$

Ex 2.2 V : a p -adically complete valuation ring

s.t. $\text{Frac}(V)$ is alg closed

$\Rightarrow V$ is perfectoid ($\varpi = p^{\frac{1}{p}}$)

(e.g., $V = \mathcal{O}_{\mathbb{C}_p}$)

Def 2.3 (Ishino-Nakazato-Shimomoto '25)

An inductive system of rings (indexed by $\mathbb{N}$)

$$R = R_0 \rightarrow R_1 \rightarrow R_2 \rightarrow \dots$$

is a **perfectoid tower arising** (from (R, p))

$\Leftrightarrow \forall i \geq 0,$
def

① R_i is p -adically complete. (for simplicity)

② $\exists \varpi \in R_i$ with $\varpi^p R_i = p R_i$

$$\begin{array}{ccc} R_{i+1}/p & \xrightarrow{\varphi} & R_{i+1}/p \\ \downarrow & \searrow \exists F_i & \uparrow \text{inj}_i \\ R_{i+1}/\varpi & \xrightarrow{\sim} & R_i/p \end{array}$$

s.t.

③ • $R_i[p^\infty] = R_i[p]$

$$\begin{array}{ccc} R_{i+1}[p] & \rightarrow & R_{i+1}/p \\ \cong \downarrow & \circlearrowleft & \downarrow F_i \\ R_i[p] & \rightarrow & R_i/p \end{array}$$

• $\widehat{R_\infty} := \left(\varinjlim_i R_i \right)_{(p)}^\wedge$ is a perfectoid ring.

• $R_i^{s.b.} \stackrel{\text{def}}{=} \varprojlim \left(R_i/p \xleftarrow{F_i} R_{i+1}/p \xleftarrow{F_{i+1}} R_{i+2}/p \xleftarrow{F_{i+2}} \dots \right)$

: i^{th} small tilt

$$\xrightarrow{(\bar{i}=0)} \text{pr}_0: R^{s.l.} \rightarrow R/p \text{ induces } R^{s.l.}/_{p^{s.l.}} \cong R/p$$

Ex 2.4 ① As in $\mathbb{Z}_p$,

every complete local log-regular ring admits a perfectoid tower.

② A : perfectoid ring

$\Rightarrow A \xrightarrow{\text{id}} A \xrightarrow{\text{id}} A \xrightarrow{\text{id}} \dots$ is a perfectoid tower.

③ (Ishizuka '26, Ishiro-Shimamoto '25)

(A, ϕ, d) : a "suitable" prism

$$\left(\begin{array}{l} \bullet \phi: \text{Frobenius lift} \\ \bullet d \in A \end{array} \right)$$

$$\begin{array}{ccc} A & \xrightarrow{\phi} & A \\ \downarrow \wr & & \downarrow \\ A/p & \xrightarrow{\varphi} & A/p \end{array}$$

$\Rightarrow A/d$ admits a perfectoid tower.

e.g., $A = \mathbb{Z}_p[\![\xi]\!] \curvearrowright \phi: \xi \mapsto \xi^p$

$$d = [p]_{\xi} = \frac{\xi^p - 1}{\xi - 1}$$

$$\Rightarrow \mathbb{Z}_p \hookrightarrow \mathbb{Z}_p[\zeta_p] \hookrightarrow \mathbb{Z}_p[\zeta_{p^2}] \hookrightarrow \dots$$

§ 3. Idea of proof

Key result (Bhatt-Mathew '21)

$\mathrm{RT}_{\text{ét}}(-, G)$ satisfies **arc** (hyper)descent.

Here $f: Y \rightarrow X$ is an $\overset{\text{(p-complete)}}{\text{arc cover}}$

$$\begin{array}{ccc} \text{def } \Leftrightarrow & \begin{array}{ccc} \exists \text{ Spec } W & \xrightarrow{\exists} & Y \\ \exists \downarrow & \curvearrowright & \downarrow f \\ \forall \text{ Spec } V & \longrightarrow & X \end{array} & \left(\begin{array}{l} \text{(p-complete)} \\ V, W: \text{a valuation ring} \\ \text{of height } \leq 1. \end{array} \right) \end{array}$$

This is the case if f : faithfully flat
or proper surj.

Const For any ring A ,

$A \rightarrow \prod_{\mathbb{P} \in \text{Spec } A} V$ is a **p-complete** arc cover

$\mathbb{P} \in \text{Spec } A$

$A \subset V \subset \overline{k(\mathbb{P})}$

p-complete val. ring of ht. ≤ 1 .

V : perfectoid
(Ex. 2.2)

$R \rightarrow \widehat{R}_{\infty} =: A \rightarrow \prod V : \text{(p-complete) arc cover}$

$\Downarrow$
 $R^{s.b.} \rightarrow \widehat{R}_{\infty}^{s.b.} = A^b \rightarrow \prod V^b : \text{(p}^{s.b.}\text{-complete) arc cover}$

$\rightarrow$ One can reduce to the case $R=V$. //